

Cell Biochemistry & Cell Signaling

60 High-Yield Multiple Choice Questions with Comprehensive Choice-by-Choice Explanations

Series: Medical Biochemistry & Cell Biology Practice Bank

Coverage: 1. Cells & Biological Membranes (Q1–20) | 2. Cellular Organelles & Biochemical Roles (Q21–40) | 3. Cell Signaling & Signal Transduction (Q41–60)

Format: Question Stem → 4 Options → Highlighted Correct Answer → Option-by-Option Rationales (A, B, C, D)

Grounding Source: *Lieberman's Marks' Basic Medical Biochemistry & Harvey & Ferrier's Biochemistry*

SECTION 1: CELLS & BIOLOGICAL MEMBRANES

Question 1

Cells & Biological Membranes

According to the fluid mosaic model of biological membranes, which component is primarily responsible for forming the continuous hydrophobic barrier?

(A) Integral transmembrane proteins

(B) Phospholipid bilayer ✓ [Correct Answer]

(C) Glycoprotein oligosaccharide chains

(D) Peripheral membrane proteins

CORRECT ANSWER: (B) Phospholipid bilayer

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Integral membrane proteins span the membrane to perform transport, receptor, or enzymatic functions, but do not form the continuous structural matrix of the membrane barrier.
- **Option B (CORRECT):** Correct. The phospholipid bilayer forms the continuous hydrophobic barrier of biological membranes, with its hydrophobic fatty acyl tails facing inward away from water and hydrophilic head groups facing the aqueous extracellular and intracellular environments.
- **Option C (Incorrect):** Incorrect. Oligosaccharide chains on glycoproteins project exclusively into the extracellular space to form the glycocalyx involved in cell recognition and adhesion, not the membrane barrier.
- **Option D (Incorrect):** Incorrect. Peripheral membrane proteins bind non-covalently to membrane surfaces and do not penetrate the hydrophobic core or form the continuous barrier.

Question 2

Cells & Biological Membranes

Which lipid component functions as a membrane 'fluidity buffer' by restricting fatty acid chain movement at high temperatures while preventing tight packing at low temperatures?

(A) Phosphatidylcholine

(B) Sphingomyelin

(C) Cholesterol ✓ [Correct Answer]

(D) Cardiolipin

CORRECT ANSWER: (C) Cholesterol

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Phosphatidylcholine is a structural glycerophospholipid, but it does not act as a bidirectional thermal fluidity buffer like cholesterol.
- **Option B (Incorrect):** Incorrect. Sphingomyelin contains long, saturated acyl chains that contribute to rigid lipid rafts, but lacks the bidirectional fluidity-buffering property of cholesterol.
- **Option C (CORRECT):** Correct. Cholesterol intercalates between membrane phospholipids. Its rigid steroid ring structure restricts excessive fatty acid movement at physiological/high temperatures, while its presence prevents hydrophobic tails from packing tightly into a rigid gel state at low temperatures.
- **Option D (Incorrect):** Incorrect. Cardiolipin is a specialized double-phospholipid found almost exclusively in the inner mitochondrial membrane to stabilize respiratory chain complexes.

Question 3

Cells & Biological Membranes

Lipid rafts are specialized microdomains in the plasma membrane enriched in cholesterol and which type of membrane lipid?

(A) Glycerophospholipids with polyunsaturated fatty acids

(B) Sphingolipids (such as sphingomyelin and glycosphingolipids) ✓ [Correct Answer]

(C) Phosphatidylinositol 4,5-bisphosphate (PIP2)

(D) Cardiolipin

CORRECT ANSWER: (B) Sphingolipids (such as sphingomyelin and glycosphingolipids)

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Polyunsaturated glycerophospholipids enhance fluid phase dynamics and are excluded from ordered lipid rafts.
- **Option B (CORRECT):** Correct. Lipid rafts are detergent-resistant microdomains enriched in cholesterol and sphingolipids (sphingomyelin and glycosphingolipids). The long, saturated hydrocarbon chains of sphingolipids pack tightly with cholesterol to form an ordered liquid phase (Lo) that scaffolds signal transduction complexes.
- **Option C (Incorrect):** Incorrect. PIP2 is an inner leaflet signaling lipid, but lipid rafts are specifically characterized by the lipid coupling of cholesterol and outer-leaflet sphingolipids.
- **Option D (Incorrect):** Incorrect. Cardiolipin is localized in the inner mitochondrial membrane, not plasma membrane lipid rafts.

Question 4*Cells & Biological Membranes*

Which amino acid residues are most commonly found in the transmembrane alpha-helical segments of integral membrane proteins?

- (A) Lysine and Arginine
- (B) Glutamate and Aspartate
- (C) **Leucine, Isoleucine, Valine, and Phenylalanine ✓ [Correct Answer]**
- (D) Serine and Threonine

CORRECT ANSWER: (C) Leucine, Isoleucine, Valine, and Phenylalanine

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Lysine and arginine are positively charged basic amino acids found at the aqueous membrane boundaries (flanking regions), not within the hydrophobic core.
- **Option B (Incorrect):** Incorrect. Glutamate and aspartate are negatively charged acidic amino acids that are thermodynamically unfavorable inside the hydrophobic lipid bilayer.
- **Option C (CORRECT):** Correct. Transmembrane alpha-helices typically consist of 20–25 nonpolar, hydrophobic amino acids—such as Leucine, Isoleucine, Valine, Phenylalanine, and Alanine—whose side chains interact favorably with the fatty acyl chains of the lipid bilayer.
- **Option D (Incorrect):** Incorrect. Serine and threonine are polar uncharged residues commonly found in aqueous loop domains or lining hydrophilic channels rather than hydrophobic transmembrane helices.

Question 5*Cells & Biological Membranes*

How are GPI (glycosylphosphatidylinositol)-anchored proteins attached to the plasma membrane?

- (A) **Covalently linked to a glycolipid on the outer leaflet of the plasma membrane ✓ [Correct Answer]**
- (B) Covalently bound to fatty acids on the cytosolic inner leaflet
- (C) Non-covalently bound to integral transmembrane proteins
- (D) Trapped in the hydrophobic core via a hydrophobic transmembrane helix

CORRECT ANSWER: (A) Covalently linked to a glycolipid on the outer leaflet of the plasma membrane

Detailed Choice-by-Choice Rationales:

- **Option A (CORRECT):** Correct. GPI anchors consist of a phosphatidylinositol moiety attached to a phosphoethanolamine-glycan chain that covalently links the C-terminus of proteins exclusively to the outer leaflet of the plasma membrane.
- **Option B (Incorrect):** Incorrect. Cytosolic inner leaflet lipid modifications include myristoylation, palmitoylation, and prenylation, not GPI anchoring.
- **Option C (Incorrect):** Incorrect. Peripheral proteins bind non-covalently, but GPI-anchored proteins are covalently tethered to membrane lipids.
- **Option D (Incorrect):** Incorrect. GPI-anchored proteins lack a hydrophobic transmembrane peptide sequence; their membrane association is entirely lipid-mediated.

Question 6

Cells & Biological Membranes

Which of the following transport mechanisms requires cellular ATP hydrolysis directly by the transport protein to move solute against its electrochemical gradient?

- (A) Facilitated diffusion via GLUT2
- (B) Primary active transport via Na⁺/K⁺-ATPase ✓ [Correct Answer]**
- (C) Secondary active transport via Na⁺/glucose cotransporter (SGLT1)
- (D) Simple diffusion through aquaporins

CORRECT ANSWER: (B) Primary active transport via Na⁺/K⁺-ATPase**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. GLUT2 operates via passive facilitated diffusion down a concentration gradient without energy expenditure.
- **Option B (CORRECT):** Correct. Primary active transport proteins (such as the Na⁺/K⁺-ATPase) directly hydrolyze ATP to drive solute transport (3 Na⁺ out, 2 K⁺ in) against steep electrochemical gradients.
- **Option C (Incorrect):** Incorrect. SGLT1 uses secondary active transport, deriving energy from the pre-existing Na⁺ electrochemical gradient maintained by the Na⁺/K⁺-ATPase rather than directly hydrolyzing ATP itself.
- **Option D (Incorrect):** Incorrect. Aquaporins facilitate water movement via passive simple/facilitated diffusion down osmotic gradients.

Question 7

Cells & Biological Membranes

The Na⁺/K⁺-ATPase pump transports ions across the plasma membrane in which stoichiometry and direction per hydrolyzed ATP molecule?

- (A) 2 Na⁺ out of the cell and 3 K⁺ into the cell
- (B) 3 Na⁺ out of the cell and 2 K⁺ into the cell ✓ [Correct Answer]**
- (C) 3 Na⁺ into the cell and 2 K⁺ out of the cell
- (D) 2 Na⁺ into the cell and 3 K⁺ out of the cell

CORRECT ANSWER: (B) 3 Na⁺ out of the cell and 2 K⁺ into the cell**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. The pump moves 3 Na⁺ out and 2 K⁺ in, not the reverse stoichiometry.
- **Option B (CORRECT):** Correct. For each ATP molecule hydrolyzed, the P-type Na⁺/K⁺-ATPase exports 3 Na⁺ ions out of the cell and imports 2 K⁺ ions into the cytosol, generating an electrogenic, net-positive outward current.
- **Option C (Incorrect):** Incorrect. Transporting Na⁺ into the cell would dissipate the physiological sodium gradient required for secondary transport.
- **Option D (Incorrect):** Incorrect. This represents inverted stoichiometry and direction.

Question 8

Cells & Biological Membranes

Glucose transport into intestinal epithelial cells via SGLT1 is an example of which transport mechanism?

- (A) Primary active transport
- (B) Sodium-dependent secondary active symport ✓ [Correct Answer]**
- (C) Facilitated diffusion uniport
- (D) Passive antiport transport

CORRECT ANSWER: (B) Sodium-dependent secondary active symport**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. SGLT1 does not hydrolyze ATP directly; primary active transport is performed by the basolateral Na⁺/K⁺-ATPase.
- **Option B (CORRECT):** Correct. SGLT1 is a secondary active symporter that couples the energetically favorable influx of 2 Na⁺ ions down their electrochemical gradient with the simultaneous uptake of 1 glucose molecule against its concentration gradient across the apical membrane.
- **Option C (Incorrect):** Incorrect. Facilitated diffusion uniport (such as GLUT2 on the basolateral membrane) moves glucose down its concentration gradient without coupling to sodium.
- **Option D (Incorrect):** Incorrect. SGLT1 transports both Na⁺ and glucose in the same direction into the cell (symport), not opposite directions (antiport).

Question 9

Cells & Biological Membranes

Which member of the GLUT transporter family is insulin-dependent and expressed primarily in adipose tissue and skeletal muscle?

- (A) GLUT1
- (B) GLUT2
- (C) GLUT3
- (D) GLUT4 ✓ [Correct Answer]**

CORRECT ANSWER: (D) GLUT4**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. GLUT1 is insulin-independent and constitutively expressed in erythrocytes, endothelial cells of the blood-brain barrier, and fetal tissues.
- **Option B (Incorrect):** Incorrect. GLUT2 is a high-K_m, low-affinity insulin-independent transporter present in liver, pancreatic beta cells, and kidney.
- **Option C (Incorrect):** Incorrect. GLUT3 is a low-K_m, high-affinity insulin-independent glucose transporter primary to neurons.
- **Option D (CORRECT):** Correct. GLUT4 is the high-affinity insulin-responsive glucose transporter stored in intracellular vesicles in striated muscle (skeletal and cardiac) and adipocytes. Insulin binding triggers GLUT4 vesicle translocation to the cell surface.

Question 10*Cells & Biological Membranes*

Cystic Fibrosis is caused by mutations in the CFTR gene. CFTR belongs to which major transport protein superfamily and functions as a transporter/channel for which ion?

- (A) P-type ATPase superfamily; Na⁺ channel
- (B) ABC (ATP-Binding Cassette) transporter family; Cl⁻ channel ✓ [Correct Answer]**
- (C) V-type ATPase family; H⁺ pump
- (D) F-type ATPase family; Ca²⁺ channel

CORRECT ANSWER: (B) ABC (ATP-Binding Cassette) transporter family; Cl⁻ channel

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. P-type ATPases form phosphorylated intermediates (e.g., Na⁺/K⁺-ATPase, SERCA) and do not include CFTR.
- **Option B (CORRECT):** Correct. CFTR (Cystic Fibrosis Transmembrane Conductance Regulator) is a unique member of the ABC transporter superfamily that functions as an ATP-gated chloride (Cl⁻) channel across epithelial cell membranes.
- **Option C (Incorrect):** Incorrect. V-type ATPases acidify organelles (lysosomes, endosomes) by pumping protons (H⁺), not Cl⁻.
- **Option D (Incorrect):** Incorrect. F-type ATPases (ATP synthase) synthesize ATP in mitochondria; they are not epithelial ion channels.

Question 11*Cells & Biological Membranes*

Which type of cell junction creates a watertight seal between adjacent epithelial cells, preventing paracellular diffusion of solutes?

- (A) Desmosomes (macula adherens)
- (B) Gap junctions (nexus)
- (C) Tight junctions (zonula occludens) ✓ [Correct Answer]**
- (D) Adherens junctions (zonula adherens)

CORRECT ANSWER: (C) Tight junctions (zonula occludens)

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Desmosomes provide mechanical resistance against shear stress by anchoring intermediate filaments, but permit fluid flux around cells.
- **Option B (Incorrect):** Incorrect. Gap junctions allow metabolic and electrical communication between adjacent cytoplasm via connexons.
- **Option C (CORRECT):** Correct. Tight junctions (zonula occludens), composed of claudin and occludin proteins, fuse adjacent cell membranes near the apical domain to seal the paracellular pathway and maintain cell polarity.
- **Option D (Incorrect):** Incorrect. Adherens junctions connect actin filament bundles between adjacent cells to mediate mechanical adhesion, not a tight barrier seal.

Question 12

Cells & Biological Membranes

Gap junctions are composed of hexameric ring assemblies of which transmembrane protein?**(A) Connexin** ✓ [Correct Answer]

(B) Cadherin

(C) Integrin

(D) Claudin

CORRECT ANSWER: (A) Connexin**Detailed Choice-by-Choice Rationales:**

- **Option A (CORRECT):** Correct. Gap junctions consist of six **connexin** protein subunits assembled into a transmembrane cylinder called a **connexon**. Two aligned connexons from neighboring cells form a hydrophilic channel allowing ions and small signaling molecules (<1 kDa) to pass.
- **Option B (Incorrect):** Incorrect. Cadherins are calcium-dependent cell adhesion molecules forming desmosomes and adherens junctions.
- **Option C (Incorrect):** Incorrect. Integrins are transmembrane receptors that anchor cells to the extracellular matrix (focal adhesions and hemidesmosomes).
- **Option D (Incorrect):** Incorrect. Claudins are structural transmembrane proteins of tight junctions.

Question 13

Cells & Biological Membranes

P-glycoprotein (MDR1) is a clinically significant efflux transporter involved in cancer drug resistance. To which protein class does it belong?

(A) Ionotropic receptor channel

(B) ATP-Binding Cassette (ABC) transporter ✓ [Correct Answer]

(C) Proton-coupled antiporter

(D) Aquaporin pore

CORRECT ANSWER: (B) ATP-Binding Cassette (ABC) transporter**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Ionotropic receptors are ligand-gated ion channels, not drug efflux pumps.
- **Option B (CORRECT):** Correct. P-glycoprotein (MDR1 / ABCB1) is an ATP-driven primary active transporter belonging to the ABC transporter family. It actively pumps hydrophobic chemotherapeutic drugs out of tumor cells, conferring multidrug resistance.
- **Option C (Incorrect):** Incorrect. MDR1 utilizes direct ATP hydrolysis rather than proton gradients.
- **Option D (Incorrect):** Incorrect. Aquaporins selectively conduct water molecules.

Question 14

Cells & Biological Membranes

Which extracellular matrix cell junction anchors epithelial cells to the underlying basement membrane via keratin intermediate filaments?

- (A) Desmosome
- (B) Hemidesmosome ✓ [Correct Answer]**
- (C) Focal adhesion
- (D) Tight junction

CORRECT ANSWER: (B) Hemidesmosome**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Desmosomes anchor keratin filaments between cell-to-cell contacts, not to the extracellular basement membrane.
- **Option B (CORRECT):** Correct. Hemidesmosomes anchor the basal surface of epithelial cells to the extracellular matrix (laminin in the basement membrane) using $\alpha 6$ - $\beta 4$ integrins linked intracellularly to keratin intermediate filaments.
- **Option C (Incorrect):** Incorrect. Focal adhesions anchor cells to ECM collagen/fibronectin using integrins linked intracellularly to actin microfilaments.
- **Option D (Incorrect):** Incorrect. Tight junctions join apical membranes of neighboring cells.

Question 15

Cells & Biological Membranes

Receptor-mediated endocytosis of LDL particles requires cytosolic coating of invaginating pits by which triskelion-structured protein?

- (A) Caveolin
- (B) Clathrin ✓ [Correct Answer]**
- (C) Dynamin
- (D) COP-II

CORRECT ANSWER: (B) Clathrin**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Caveolin forms flask-shaped caveolae microdomains involved in transcytosis and potocytosis, not classical LDL receptor endocytosis.
- **Option B (CORRECT):** Correct. Clathrin molecules consist of three heavy and three light chains forming a triskelion structure. Clathrin triskelions assemble into a polyhedral lattice on the cytosolic face of plasma membrane pits to drive receptor-mediated endocytosis of cargo like LDL.
- **Option C (Incorrect):** Incorrect. Dynamin is a GTPase enzyme that pinches off the neck of the invaginated clathrin-coated pit.
- **Option D (Incorrect):** Incorrect. COP-II coatomer proteins mediate anterograde vesicle transport from the endoplasmic reticulum to the Golgi apparatus.

Question 16

Cells & Biological Membranes

Digitalis cardiac glycosides (such as digoxin) increase cardiac contractility primarily by inhibiting which membrane transport protein?

- (A) Sarco/endoplasmic reticulum Ca^{2+} -ATPase (SERCA)
- (B) Plasma membrane $\text{Na}^{+}/\text{K}^{+}$ -ATPase ✓ [Correct Answer]**
- (C) $\text{Na}^{+}/\text{Ca}^{2+}$ exchanger (NCX)
- (D) L-type voltage-gated Ca^{2+} channels

CORRECT ANSWER: (B) Plasma membrane $\text{Na}^{+}/\text{K}^{+}$ -ATPase**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Digoxin does not target SERCA pumps directly.
- **Option B (CORRECT):** Correct. Digoxin inhibits the extracellular K^{+} -binding site of the cardiac sarcolemmal $\text{Na}^{+}/\text{K}^{+}$ -ATPase. This increases intracellular $[\text{Na}^{+}]$, which reduces the driving force for the $\text{Na}^{+}/\text{Ca}^{2+}$ exchanger (NCX), leading to elevated intracellular $[\text{Ca}^{2+}]$ and enhanced cardiac inotropy.
- **Option C (Incorrect):** Incorrect. Digoxin indirectly diminishes NCX activity due to intracellular Na^{+} accumulation, but directly binds and inhibits $\text{Na}^{+}/\text{K}^{+}$ -ATPase.
- **Option D (Incorrect):** Incorrect. Calcium channel blockers (e.g., verapamil) inhibit L-type calcium channels.

Question 17

Cells & Biological Membranes

Which lipid asymmetry feature is a classic early marker of apoptosis when translocated from the inner cytosolic leaflet to the outer extracellular leaflet of the plasma membrane?

- (A) Phosphatidylcholine
- (B) Phosphatidylserine ✓ [Correct Answer]**
- (C) Sphingomyelin
- (D) Phosphatidylinositol

CORRECT ANSWER: (B) Phosphatidylserine**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Phosphatidylcholine is normally concentrated on the outer leaflet.
- **Option B (CORRECT):** Correct. Under physiological conditions, the ATP-dependent flippase keeps phosphatidylserine restricted to the cytosolic inner leaflet. During apoptosis, scramblase activation and flippase inactivation translocate phosphatidylserine to the outer leaflet, serving as an 'eat-me' signal for phagocytes.
- **Option C (Incorrect):** Incorrect. Sphingomyelin resides naturally on the outer extracellular leaflet.
- **Option D (Incorrect):** Incorrect. Phosphatidylinositol is an inner-leaflet signaling phospholipid precursor, but outer-leaflet exposure of phosphatidylserine is the specific apoptotic hallmark.

Question 18

Cells & Biological Membranes

Which enzyme family utilizes ATP hydrolysis to specifically move aminophospholipids (phosphatidylserine and phosphatidylethanolamine) from the extracellular outer leaflet to the cytosolic inner leaflet?

(A) Floppases (ABC transporters)

(B) Flippases (P4-type ATPases) ✓ [Correct Answer]

(C) Scramblases

(D) Phospholipase C

CORRECT ANSWER: (B) Flippases (P4-type ATPases)

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Floppases are ABC transporters that move lipids outward from the inner leaflet to the outer leaflet against concentration gradients.
- **Option B (CORRECT):** Correct. Flippases (P4-type ATPases) selectively transport inward, moving aminophospholipids (phosphatidylserine and phosphatidylethanolamine) from the outer leaflet to the cytosolic inner leaflet using ATP hydrolysis.
- **Option C (Incorrect):** Incorrect. Scramblases are ATP-independent, Ca^{2+} -activated transporters that non-specifically randomize phospholipids across both leaflets.
- **Option D (Incorrect):** Incorrect. Phospholipase C is a cleavage enzyme hydrolyzing PIP₂ into IP₃ and DAG.

Question 19

Cells & Biological Membranes

Which type of passive transport protein selectively conducts water molecules across biological membranes at high velocities without allowing proton passage?

(A) Aquaporin ✓ [Correct Answer]

(B) Gramicidin channel

(C) Connexon

(D) Ionophore

CORRECT ANSWER: (A) Aquaporin

Detailed Choice-by-Choice Rationales:

- **Option A (CORRECT):** Correct. Aquaporins (e.g., AQP1, AQP2) form narrow transmembrane channels that conduct water in single file while preventing proton (H_3O^+) hopping via steric constraints and conserved asparagine-proline-alanine (NPA) motifs.
- **Option B (Incorrect):** Incorrect. Gramicidin is an antibiotic peptide forming non-selective monovalent cation channels.
- **Option C (Incorrect):** Incorrect. Connexons form wider gap junction pores passing water, ions, and small metabolites up to 1 kDa.
- **Option D (Incorrect):** Incorrect. Ionophores transport specific inorganic ions across lipid membranes.

Question 20

Cells & Biological Membranes

Cholera toxin induces severe secretory diarrhea by ADP-ribosylating the Gs alpha subunit. This leads to constitutive activation of adenyl cyclase and opening of which intestinal apical membrane channel?

(A) ENaC (Epithelial Sodium Channel)

(B) CFTR (Cystic Fibrosis Transmembrane Conductance Regulator Cl⁻ channel) ✓ [Correct Answer]

(C) SGLT1 transporter

(D) Na⁺/K⁺-ATPase pump

CORRECT ANSWER: (B) CFTR (Cystic Fibrosis Transmembrane Conductance Regulator Cl⁻ channel)

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. ENaC reabsorbs Na⁺; its activation would increase sodium absorption rather than massive Cl⁻ and water secretion.
- **Option B (CORRECT):** Correct. Elevated cAMP produced by cholera toxin-mediated persistent Gs activation hyperactivates Protein Kinase A (PKA). PKA phosphorylates the R-domain of CFTR, opening the apical Cl⁻ channel. Massive Cl⁻ efflux into the lumen drives sodium and water secretion, resulting in watery diarrhea.
- **Option C (Incorrect):** Incorrect. SGLT1 is an apical glucose/Na⁺ cotransporter utilized in oral rehydration therapy to restore fluid uptake.
- **Option D (Incorrect):** Incorrect. Na⁺/K⁺-ATPase is located on the basolateral membrane.

SECTION 2: CELLULAR ORGANELLES

Question 21

Cellular Organelles

Which sub-nuclear structure is the primary site of ribosomal RNA (rRNA) transcription, processing, and ribosome subunit assembly?

(A) Nuclear matrix

(B) Nucleolus ✓ [Correct Answer]

(C) Cajal body

(D) Nuclear envelope

CORRECT ANSWER: (B) Nucleolus

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. The nuclear matrix is a structural framework supporting chromatin organization, not the primary rRNA ribosome assembly site.
- **Option B (CORRECT):** Correct. The nucleolus is a non-membrane-bound nuclear compartment where RNA Polymerase I transcribes precursor 45S rRNA, which is processed and assembled with ribosomal proteins into 40S and 60S subunit precursors.
- **Option C (Incorrect):** Incorrect. Cajal bodies are involved in snRNA and snoRNA maturation, not ribosome subunit assembly.
- **Option D (Incorrect):** Incorrect. The nuclear envelope is a double membrane enclosing the nucleoplasm containing nuclear pore complexes.

Question 22

Cellular Organelles

Nuclear localization signals (NLS) on cytosolic proteins bound for nuclear import are recognized by which nuclear transport receptor family?

(A) Importins (Karyopherins) ✓ [Correct Answer]

(B) Exportins

(C) Signal Recognition Particle (SRP)

(D) Clathrin coatomers

CORRECT ANSWER: (A) Importins (Karyopherins)

Detailed Choice-by-Choice Rationales:

- **Option A (CORRECT):** Correct. Proteins containing a classic basic Nuclear Localization Signal (NLS) are bound in the cytosol by **Importin- α /beta** heterodimers. Importins dock at nuclear pore complexes (NPC) and mediate translocation into the nucleus.
- **Option B (Incorrect):** Incorrect. Exportins mediate nuclear export of proteins containing Nuclear Export Signals (NES) in complex with Ran-GTP.
- **Option C (Incorrect):** Incorrect. SRP targets ribosome-nascent chain complexes to the endoplasmic reticulum membrane.
- **Option D (Incorrect):** Incorrect. Clathrin coats endocytic vesicles at the plasma membrane and trans-Golgi network.

Question 23

Cellular Organelles

Small GTPase Ran regulates directional nucleocytoplasmic transport. Which enzyme is concentrated in the nucleus to maintain Ran in its GTP-bound state?

(A) Ran-GAP (GTPase Activating Protein)

(B) Ran-GEF (Guanine Nucleotide Exchange Factor / RCC1) ✓ [Correct Answer]

(C) Protein Kinase A

(D) Phospholipase C

CORRECT ANSWER: (B) Ran-GEF (Guanine Nucleotide Exchange Factor / RCC1)

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Ran-GAP is localized in the cytosol, promoting Ran GTP hydrolysis (Ran-GTP $\rightarrow$ Ran-GDP).
- **Option B (CORRECT):** Correct. **Ran-GEF (RCC1)** is bound to chromatin inside the nucleus. It catalyzes GDP dissociation and GTP loading, establishing a high concentration of **Ran-GTP in the nucleus**, which drives cargo release from importins and cargo binding to exportins.
- **Option C (Incorrect):** Incorrect. PKA is a cAMP-dependent kinase, not a Ran nucleotide exchange factor.
- **Option D (Incorrect):** Incorrect. Phospholipase C generates second messengers IP3 and DAG.

Question 24

Cellular Organelles

The inner mitochondrial membrane is uniquely enriched in which phospholipid that contains four fatty acyl chains and is required for optimal respiratory complex activity?

- (A) Phosphatidylinositol
- (B) Sphingomyelin
- (C) **Cardiolipin (Diphosphatidylglycerol) ✓ [Correct Answer]**
- (D) Phosphatidylserine

CORRECT ANSWER: (C) Cardiolipin (Diphosphatidylglycerol)**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Phosphatidylinositol is predominantly in plasma membrane leaflets.
- **Option B (Incorrect):** Incorrect. Sphingomyelin is abundant in plasma membrane lipid rafts.
- **Option C (CORRECT):** Correct. **Cardiolipin** constitutes ~20% of inner mitochondrial membrane lipids. Its unique structure (two phosphatidic acids linked by glycerol) creates tight membrane curvature and stabilizes Electron Transport Chain complexes (Complexes I, III, IV) and ATP synthase.
- **Option D (Incorrect):** Incorrect. Phosphatidylserine is cytosolic inner leaflet lipid.

Question 25

Cellular Organelles

Which of the following metabolic pathways occurs exclusively within the mitochondrial matrix?

- (A) Glycolysis
- (B) Pentose Phosphate Pathway
- (C) **TCA (Citric Acid) Cycle ✓ [Correct Answer]**
- (D) Fatty Acid De Novo Synthesis

CORRECT ANSWER: (C) TCA (Citric Acid) Cycle**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Glycolysis takes place entirely in the cytosol.
- **Option B (Incorrect):** Incorrect. The Pentose Phosphate Pathway is cytosolic.
- **Option C (CORRECT):** Correct. The **TCA cycle** (as well as fatty acid beta-oxidation and pyruvate dehydrogenase conversion) occurs inside the mitochondrial matrix (except succinate dehydrogenase/Complex II which is inner membrane-bound).
- **Option D (Incorrect):** Incorrect. Fatty acid synthesis occurs in the cytosol.

Question 26

Cellular Organelles

Proteins destined for secretion or lysosomes are translocated into the Rough Endoplasmic Reticulum (RER) lumen during translation. What recognizes the hydrophobic N-terminal signal peptide?

(A) Signal Recognition Particle (SRP) ✓ [Correct Answer]

(B) Chaperone BiP/Hsp70

(C) Ubiquitin ligase

(D) COP-I coatomer

CORRECT ANSWER: (A) Signal Recognition Particle (SRP)

Detailed Choice-by-Choice Rationales:

- **Option A (CORRECT):** Correct. As a hydrophobic signal peptide emerges from the ribosome, the cytosolic **Signal Recognition Particle (SRP)** binds the peptide, halts translation, and docks the ribosome-nascent chain complex to the SRP receptor on the RER membrane.
- **Option B (Incorrect):** Incorrect. BiP is an luminal RER chaperone that assists in folding inside the lumen after translocation.
- **Option C (Incorrect):** Incorrect. Ubiquitin ligases target cytosolic or misfolded ERAD proteins for proteasomal degradation.
- **Option D (Incorrect):** Incorrect. COP-I coatomers mediate retrograde transport from Golgi back to ER.

Question 27

Cellular Organelles

Initial N-linked glycosylation of proteins in the RER involves transferring a preformed 14-sugar oligosaccharide precursor from which membrane-bound lipid carrier?

(A) Dolichol phosphate ✓ [Correct Answer]

(B) Ceramide

(C) Cholesterol

(D) Squalene

CORRECT ANSWER: (A) Dolichol phosphate

Detailed Choice-by-Choice Rationales:

- **Option A (CORRECT):** Correct. **Dolichol phosphate**, a specialized isoprenoid lipid in the ER membrane, serves as the anchor upon which the branched 14-sugar oligosaccharide (Glc3Man9GlcNAc2) is preassembled before transfer to an asparagine residue (Asn-X-Ser/Thr) by oligosaccharyltransferase.
- **Option B (Incorrect):** Incorrect. Ceramide is the sphingolipid backbone for sphingomyelin and glycosphingolipids synthesized in the Golgi.
- **Option C (Incorrect):** Incorrect. Cholesterol is a structural steroid lipid.
- **Option D (Incorrect):** Incorrect. Squalene is a cytosolic/ER intermediate in cholesterol synthesis.

Question 28

Cellular Organelles

Which organelle is the primary site of steroid hormone synthesis, phospholipid biogenesis, drug detoxification via Cytochrome P450, and intracellular Ca²⁺ storage?

- (A) Rough Endoplasmic Reticulum
- (B) Smooth Endoplasmic Reticulum ✓ [Correct Answer]**
- (C) Golgi Apparatus
- (D) Peroxisome

CORRECT ANSWER: (B) Smooth Endoplasmic Reticulum

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Rough ER is studded with ribosomes and specializes in secretory/membrane protein translation and N-glycosylation.
- **Option B (CORRECT):** Correct. The **Smooth Endoplasmic Reticulum (SER)** lacks ribosomes and houses enzymes for cholesterol/steroid synthesis, lipid biogenesis, xenobiotic hydroxylation/detoxification by Cytochrome P450 enzymes, and sequestration of Ca²⁺ ions (e.g., sarcoplasmic reticulum in muscle).
- **Option C (Incorrect):** Incorrect. Golgi apparatus modifies oligosaccharides and sorts proteins.
- **Option D (Incorrect):** Incorrect. Peroxisomes handle VLCFA oxidation and ROS detoxification.

Question 29

Cellular Organelles

Proteins synthesized in the ER are transported to the cis-Golgi network in vesicles coated with which protein complex?

- (A) COP-I
- (B) COP-II ✓ [Correct Answer]**
- (C) Clathrin
- (D) SNARE

CORRECT ANSWER: (B) COP-II

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. COP-I coats vesicles mediating **retrograde** transport from the cis-Golgi back to the ER.
- **Option B (CORRECT):** Correct. **COP-II** coatamer proteins assemble at ER exit sites to form vesicles for **anterograde** transport of cargo from the ER to the cis-Golgi network.
- **Option C (Incorrect):** Incorrect. Clathrin coats vesicles moving from trans-Golgi to endosomes or plasma membrane endocytosis.
- **Option D (Incorrect):** Incorrect. SNARE proteins (v-SNARE and t-SNARE) mediate vesicle membrane fusion, not vesicle coat assembly.

Question 30

Cellular Organelles

Lysosomal enzymes synthesized in the RER receive a specific carbohydrate modification in the cis-Golgi that targets them to lysosomes. What is this recognition signal?

(A) Sialic acid-6-phosphate

(B) Mannose-6-phosphate (M6P) ✓ [Correct Answer]

(C) Glucose-6-phosphate

(D) Galactose-1-phosphate

CORRECT ANSWER: (B) Mannose-6-phosphate (M6P)

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Sialic acid capping occurs on secretory glycoproteins in trans-Golgi.
- **Option B (CORRECT):** Correct. In the cis-Golgi, lysosomal acid hydrolases are recognized by UDP-GlcNAc phosphotransferase, which adds **Mannose-6-phosphate (M6P)** tags. M6P receptors in the trans-Golgi bind these tagged enzymes and package them into clathrin-coated vesicles destined for lysosomes.
- **Option C (Incorrect):** Incorrect. Glucose-6-phosphate is a glycolytic intermediate.
- **Option D (Incorrect):** Incorrect. Galactose-1-phosphate is an intermediate in galactose metabolism.

Question 31

Cellular Organelles

I-Cell disease (Mucopolipidosis Type II) is a severe lysosomal storage disorder caused by a defect in which enzyme required for lysosomal enzyme targeting?

(A) Glucocerebrosidase

(B) GlcNAc phosphotransferase (N-acetylglucosaminyl-1-phosphotransferase) ✓ [Correct Answer]

(C) Hexosaminidase A

(D) Acid sphingomyelinase

CORRECT ANSWER: (B) GlcNAc phosphotransferase (N-acetylglucosaminyl-1-phosphotransferase)

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Deficiency of glucocerebrosidase causes Gaucher disease.
- **Option B (CORRECT):** Correct. **I-Cell disease** results from mutations in the **GNPTAB** gene encoding **GlcNAc phosphotransferase**. Without this enzyme, lysosomal acid hydrolases cannot be phosphorylated to form Mannose-6-Phosphate tags. Un-tagged enzymes are misrouted into extracellular space/blood, while dense undigested inclusions ('I-cells') accumulate in lysosomes.
- **Option C (Incorrect):** Incorrect. Deficiency of Hexosaminidase A causes Tay-Sachs disease.
- **Option D (Incorrect):** Incorrect. Deficiency of acid sphingomyelinase causes Niemann-Pick disease.

Question 32

Cellular Organelles

Tay-Sachs disease is an autosomal recessive lysosomal storage disease caused by a deficiency of Hexosaminidase A, leading to neuronal accumulation of which lipid?

(A) Glucocerebroside

(B) GM2 Ganglioside ✓ [Correct Answer]

(C) Sphingomyelin

(D) Galactocerebroside

CORRECT ANSWER: (B) GM2 Ganglioside

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Glucocerebroside accumulates in Gaucher disease.
- **Option B (CORRECT):** Correct. **Tay-Sachs disease** is caused by a deficiency of **Hexosaminidase A** (alpha-subunit defect), leading to progressive toxic accumulation of **GM2 ganglioside** in lysosomes of central nervous system neurons, presenting with neurodegeneration, developmental regression, and a cherry-red macular spot.
- **Option C (Incorrect):** Incorrect. Sphingomyelin accumulates in Niemann-Pick disease.
- **Option D (Incorrect):** Incorrect. Galactocerebroside accumulates in Krabbe disease.

Question 33

Cellular Organelles

Gaucher disease, the most common lysosomal storage disorder, is caused by a deficiency of beta-glucocerebrosidase and results in lipid accumulation in macrophages ('Gaucher cells'). Which lipid accumulates?

(A) Glucocerebroside (Glucosylceramide) ✓ [Correct Answer]

(B) GM2 Ganglioside

(C) Sphingomyelin

(D) Ceramide trihexoside

CORRECT ANSWER: (A) Glucocerebroside (Glucosylceramide)

Detailed Choice-by-Choice Rationales:

- **Option A (CORRECT):** Correct. Deficiency of lysosomal **beta-glucocerebrosidase** (acid beta-glucosidase) causes **Gaucher disease**. **Glucocerebroside** accumulates in reticuloendothelial macrophages, giving cytoplasm a classic 'crinkled paper' appearance with hepatosplenomegaly, bone crises, and cytopenias.
- **Option B (Incorrect):** Incorrect. GM2 ganglioside accumulates in Tay-Sachs disease.
- **Option C (Incorrect):** Incorrect. Sphingomyelin accumulates in Niemann-Pick disease.
- **Option D (Incorrect):** Incorrect. Ceramide trihexoside accumulates in Fabry disease.

Question 34

Cellular Organelles

Which organelle performs beta-oxidation of Very Long Chain Fatty Acids (VLCFAs, >22 carbons) and synthesizes ether phospholipids (plasmalogens)?

(A) Mitochondria

(B) Peroxisome ✓ [Correct Answer]

(C) Lysosome

(D) Golgi apparatus

CORRECT ANSWER: (B) Peroxisome

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Mitochondria oxidize short, medium, and long-chain fatty acids (<20 carbons), but cannot shorten VLCFAs directly.
- **Option B (CORRECT):** Correct. **Peroxisomes** contain enzymes for initial alpha- and beta-oxidation of **Very Long Chain Fatty Acids (VLCFAs)** and branched-chain fatty acids (phytanic acid), generating H₂O₂ that catalase degrades. Peroxisomes also house DHAP acyltransferase required for plasmalogen synthesis.
- **Option C (Incorrect):** Incorrect. Lysosomes digest macromolecules via acid hydrolases.
- **Option D (Incorrect):** Incorrect. Golgi apparatus modifies carbohydrate side chains.

Question 35

Cellular Organelles

Zellweger syndrome is a lethal congenital disorder caused by mutations in PEX genes. What is the cellular basis of Zellweger syndrome?

(A) Failure of peroxisome biogenesis and protein import ✓ [Correct Answer]

(B) Defect in lysosomal acidification via V-type H⁺-ATPase

(C) Mitochondrial electron transport chain Complex I deficiency

(D) Failure of COP-I vesicle budding

CORRECT ANSWER: (A) Failure of peroxisome biogenesis and protein import

Detailed Choice-by-Choice Rationales:

- **Option A (CORRECT):** Correct. **Zellweger syndrome** (cerebrohepato-renal syndrome) is an autosomal recessive disorder caused by defective **PEX genes** (peroxins) required for peroxisomal membrane assembly and PTS1/PTS2 signal matrix protein import. Empty peroxisomal 'ghosts' cause severe VLCFA accumulation, lack of plasmalogens, hypotonia, and early death.
- **Option B (Incorrect):** Incorrect. Lysosomal acidification defects cause osteopetrosis or storage syndromes, not Zellweger.
- **Option C (Incorrect):** Incorrect. Complex I deficiency causes Leigh syndrome or MELAS.
- **Option D (Incorrect):** Incorrect. COP-I defects disrupt Golgi-to-ER retrograde transport.

Question 36

Cellular Organelles

Ubiquitin-tagged misfolded proteins are targeted for ATP-dependent degradation in which cytosolic macromolecular complex?

(A) 26S Proteasome ✓ [Correct Answer]

(B) Lysosome

(C) Peroxisome

(D) Phagolysosome

CORRECT ANSWER: (A) 26S Proteasome

Detailed Choice-by-Choice Rationales:

- **Option A (CORRECT):** Correct. Proteins marked for destruction by polyubiquitination (K48-linked chains) are recognized, unfolded, and degraded into small peptides by the **26S Proteasome** (20S catalytic core plus 19S regulatory caps).
- **Option B (Incorrect):** Incorrect. Lysosomes degrade extracellular, membrane, or organellar proteins via autophagy/endocytosis.
- **Option C (Incorrect):** Incorrect. Peroxisomes oxidize fatty acids and detoxify ROS.
- **Option D (Incorrect):** Incorrect. Phagolysosomes digest phagocytosed microorganisms.

Question 37

Cellular Organelles

Which cytoskeletal component has a diameter of ~25 nm, is composed of alpha/beta-tubulin heterodimers, and provides tracks for kinesin and dynein motor proteins?

(A) Microfilaments (Actin)

(B) Intermediate filaments (Keratin)

(C) Microtubules ✓ [Correct Answer]

(D) Myofibrils

CORRECT ANSWER: (C) Microtubules

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Microfilaments are 7 nm thin filaments composed of actin.
- **Option B (Incorrect):** Incorrect. Intermediate filaments are 8–10 nm fibers providing mechanical stability.
- **Option C (CORRECT):** Correct. **Microtubules** are hollow cylinders (~25 nm diameter) composed of 13 protofilaments of polymerized **alpha- and beta-tubulin** heterodimers. They maintain cell shape, form mitotic spindles, and serve as structural tracks for kinesin (plus-end) and dynein (minus-end) motor proteins.
- **Option D (Incorrect):** Incorrect. Myofibrils are contractile actin-myosin units in muscle cells.

Question 38

Cellular Organelles

Colchicine, Vincristine, and Paclitaxel target which cytoskeletal element to disrupt mitosis in cancer chemotherapy or gout treatment?

- (A) Actin microfilaments
- (B) Microtubules ✓ [Correct Answer]**
- (C) Vimentin intermediate filaments
- (D) Nuclear lamins

CORRECT ANSWER: (B) Microtubules**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Cytochalasins and phalloidin target actin microfilaments.
- **Option B (CORRECT):** Correct. **Colchicine** and **Vincristine/Vinblastine** bind tubulin to prevent microtubule polymerization, while **Paclitaxel (Taxol)** hyper-stabilizes microtubules to prevent depolymerization. Both mechanisms arrest cells in metaphase of mitosis by disrupting spindle dynamics.
- **Option C (Incorrect):** Incorrect. Vimentin is an intermediate filament in mesenchymal cells.
- **Option D (Incorrect):** Incorrect. Nuclear lamins form the nuclear lamina supporting the inner nuclear membrane.

Question 39

Cellular Organelles

Which molecular motor protein moves membrane-bound organelles along microtubules towards the plus-end (anterograde transport, towards the cell periphery)?

- (A) Dynein
- (B) Kinesin ✓ [Correct Answer]**
- (C) Myosin II
- (D) Cofilin

CORRECT ANSWER: (B) Kinesin**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Cytoplasmic dynein is a minus-end directed motor protein (retrograde transport towards the microtubule organizing center/centrosome).
- **Option B (CORRECT):** Correct. **Kinesin** is an ATP-dependent motor protein that walks along microtubules towards the **plus-end (+)**, driving anterograde transport of vesicles and organelles from the cell center toward the periphery.
- **Option C (Incorrect):** Incorrect. Myosin II moves along actin filaments during muscle contraction.
- **Option D (Incorrect):** Incorrect. Cofilin is an actin-severing protein, not a motor protein.

Question 40

Cellular Organelles

Nuclear Lamins are specialized cytoskeletal proteins forming the meshwork underlying the inner nuclear membrane. To which cytoskeletal class do lamins belong?

- (A) Microtubules
- (B) Actin microfilaments
- (C) Intermediate filaments ✓ [Correct Answer]
- (D) Clathrin lattices

CORRECT ANSWER: (C) Intermediate filaments**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Microtubules are tubulin polymers found in cytoplasm and mitotic spindles.
- **Option B (Incorrect):** Incorrect. Actin microfilaments form cortical actin networks.
- **Option C (CORRECT):** Correct. ****Nuclear Lamins (Lamins A, B, and C)**** are Type V ****intermediate filaments**** that assemble into a 2D fibrous meshwork (nuclear lamina) supporting the inner nuclear envelope and anchoring chromatin. Phosphorylation of lamins by CDK1 causes nuclear envelope breakdown during mitosis.
- **Option D (Incorrect):** Incorrect. Clathrin lattices coat endocytic membrane vesicles.

SECTION 3: CELL SIGNALING

Question 41

Cell Signaling

Which mode of extracellular cell signaling involves signaling molecules acting locally on neighboring target cells of a different type in the immediate extracellular environment?

- (A) Endocrine signaling
- (B) Paracrine signaling ✓ [Correct Answer]
- (C) Autocrine signaling
- (D) Juxtacrine signaling

CORRECT ANSWER: (B) Paracrine signaling**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Endocrine signaling involves hormones released into the bloodstream to act on distant target tissues.
- **Option B (CORRECT):** Correct. ****Paracrine signaling**** involves cells secreting signal molecules (such as neurotransmitters, histamine, or local growth factors) into the interstitial fluid to act on neighboring target cells in close physical proximity.
- **Option C (Incorrect):** Incorrect. Autocrine signaling occurs when a cell secretes signals that bind to receptors on its own surface.
- **Option D (Incorrect):** Incorrect. Juxtacrine signaling requires direct cell-to-cell contact via membrane-bound ligands and receptors (e.g., Notch signaling).

Question 42

Cell Signaling

G-protein coupled receptors (GPCRs) span the plasma membrane how many times?

- (A) Single transmembrane pass (1)
- (B) Seven transmembrane alpha-helices (7) ✓ [Correct Answer]**
- (C) Twelve transmembrane passes (12)
- (D) Four transmembrane passes (4)

CORRECT ANSWER: (B) Seven transmembrane alpha-helices (7)

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Single transmembrane-pass receptors include Receptor Tyrosine Kinases (RTKs) and cytokine receptors.
- **Option B (CORRECT):** Correct. **GPCRs** (also called 7-TM receptors or serpentine receptors) possess **seven hydrophobic transmembrane alpha-helical segments** connected by three extracellular loops and three intracellular loops.
- **Option C (Incorrect):** Incorrect. 12-TM structures are characteristic of solute transporters like GLUT or ABC transporters.
- **Option D (Incorrect):** Incorrect. 4-TM subunits are typical of ligand-gated ion channels like nicotinic acetylcholine receptor subunits.

Question 43

Cell Signaling

When a ligand binds to a G-protein coupled receptor, what molecular event triggers activation of the heterotrimeric G protein?

- (A) Phosphorylation of GDP to GTP on the G-beta subunit
- (B) Exchange of GDP for GTP on the G-alpha subunit ✓ [Correct Answer]**
- (C) Hydrolysis of GTP to GDP on the G-gamma subunit
- (D) Proteolytic cleavage of the G-alpha subunit

CORRECT ANSWER: (B) Exchange of GDP for GTP on the G-alpha subunit

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. GDP is exchanged for GTP, not directly phosphorylated on the beta subunit.
- **Option B (CORRECT):** Correct. Ligand binding induces a conformational change in the GPCR that acts as a Guanine Nucleotide Exchange Factor (GEF), causing the **G-alpha subunit to release GDP and bind GTP**. This triggers dissociation of G-alpha-GTP from the G-beta-gamma dimer to regulate downstream effectors.
- **Option C (Incorrect):** Incorrect. GTP hydrolysis inactivates G-alpha; it does not activate the G-protein.
- **Option D (Incorrect):** Incorrect. Activation involves nucleotide exchange, not proteolytic cleavage.

Question 44

Cell Signaling

Activation of the Gs-alpha subunit by glucagon or epinephrine stimulates which plasma membrane effector enzyme?

(A) Phospholipase C-beta

(B) Adenylyl cyclase ✓ [Correct Answer]

(C) cGMP phosphodiesterase

(D) Protein Kinase C

CORRECT ANSWER: (B) Adenylyl cyclase

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Phospholipase C-beta is activated by Gq-alpha subunits.
- **Option B (CORRECT):** Correct. **Gs-alpha-GTP** binds directly to and activates **Adenylyl cyclase**, an integral membrane enzyme that converts cytosolic ATP into the second messenger cyclic AMP (cAMP).
- **Option C (Incorrect):** Incorrect. cGMP phosphodiesterase is activated by Transducin (Gt) in retinal photoreceptors.
- **Option D (Incorrect):** Incorrect. PKC is activated by DAG and Ca²⁺ down Gq signaling pathways.

Question 45

Cell Signaling

Cyclic AMP (cAMP) activates Protein Kinase A (PKA) by which mechanism?

(A) Phosphorylating the PKA catalytic subunits directly

(B) Binding to PKA regulatory subunits and causing their dissociation from catalytic subunits ✓ [Correct Answer]

(C) Inducing dimerization of PKA transmembrane helices

(D) Cleaving PKA into active fragments

CORRECT ANSWER: (B) Binding to PKA regulatory subunits and causing their dissociation from catalytic subunits

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. cAMP does not possess kinase activity to phosphorylate PKA.
- **Option B (CORRECT):** Correct. Inactive PKA exists as a tetramer of two regulatory (R) and two catalytic (C) subunits (R₂C₂). Binding of **four cAMP molecules** (two to each R subunit) induces a conformational change that causes the **R subunits to dissociate**, releasing the active catalytic (C) monomeric kinases.
- **Option C (Incorrect):** Incorrect. PKA is a cytosolic kinase, not a transmembrane receptor.
- **Option D (Incorrect):** Incorrect. PKA activation is reversible binding, not proteolytic cleavage.

Question 46

Cell Signaling

Which G-protein alpha subunit family activates Phospholipase C-beta (PLC-beta) to cleave membrane PIP2 into IP3 and Diacylglycerol (DAG)?

(A) Gs-alpha

(B) Gi-alpha

(C) Gq-alpha ✓ [Correct Answer]

(D) Gt-alpha (Transducin)

CORRECT ANSWER: (C) Gq-alpha**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Gs-alpha stimulates adenylyl cyclase.
- **Option B (Incorrect):** Incorrect. Gi-alpha inhibits adenylyl cyclase and lowers cAMP.
- **Option C (CORRECT):** Correct. **Gq-alpha-GTP** activates **Phospholipase C-beta (PLC-beta)**. PLC-beta cleaves phosphatidylinositol 4,5-bisphosphate (PIP2) into two second messengers: inositol 1,4,5-trisphosphate (**IP3**) and 1,2-diacylglycerol (**DAG**).
- **Option D (Incorrect):** Incorrect. Gt-alpha activates cGMP phosphodiesterase.

Question 47

Cell Signaling

Inositol 1,4,5-trisphosphate (IP3) generated by PLC-beta binds to ligand-gated ion channels located on which organelle membrane to release Ca2+ into cytosol?

(A) Mitochondrial inner membrane

(B) Endoplasmic / Sarcoplasmic Reticulum ✓ [Correct Answer]

(C) Lysosome

(D) Peroxisome

CORRECT ANSWER: (B) Endoplasmic / Sarcoplasmic Reticulum**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Mitochondria buffer cytosolic Ca2+ via calcium uniporters, but IP3 receptors reside on the ER.
- **Option B (CORRECT):** Correct. **IP3** diffuses rapidly through the cytosol and binds to **IP3 receptor Ca2+ channels** on the **Endoplasmic Reticulum (or sarcoplasmic reticulum)** membrane, causing rapid efflux of stored Ca2+ into the cytoplasm.
- **Option C (Incorrect):** Incorrect. Lysosomal Ca2+ release is regulated by NAADP / TPC channels.
- **Option D (Incorrect):** Incorrect. Peroxisomes do not serve as IP3-gated Ca2+ stores.

Question 48

Cell Signaling

Diacylglycerol (DAG) produced by PIP₂ cleavage remains in the plasma membrane and, together with elevated cytosolic Ca²⁺, activates which protein kinase?

- (A) Protein Kinase A (PKA)
- (B) Protein Kinase B (Akt)
- (C) Protein Kinase C (PKC) ✓ [Correct Answer]**
- (D) Cyclin-Dependent Kinase 1 (CDK1)

CORRECT ANSWER: (C) Protein Kinase C (PKC)**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. PKA is activated by cAMP.
- **Option B (Incorrect):** Incorrect. Akt/PKB is activated downstream of PI3K and PIP₃.
- **Option C (CORRECT):** Correct. **DAG** remains embedded in the plasma membrane inner leaflet. Membrane-bound DAG together with elevated cytosolic **Ca²⁺** recruits and fully activates conventional isoforms of **Protein Kinase C (PKC)** to phosphorylate target cellular proteins.
- **Option D (Incorrect):** Incorrect. CDK1 is activated by Cyclin B to drive mitosis.

Question 49

Cell Signaling

Pertussis toxin causes Whooping Cough by modifying which G-protein subunit, preventing it from interacting with GPCRs and trapping it in an inactive state?

- (A) ADP-ribosylation of Gs-alpha
- (B) ADP-ribosylation of Gi-alpha ✓ [Correct Answer]**
- (C) Phosphorylation of Gq-alpha
- (D) Deacetylation of G12/13-alpha

CORRECT ANSWER: (B) ADP-ribosylation of Gi-alpha**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Cholera toxin ADP-ribosylates Gs-alpha to lock it in the active GTP-bound state.
- **Option B (CORRECT):** Correct. **Pertussis toxin** catalyzes **ADP-ribosylation of Gi-alpha** at a cysteine residue. This modification prevents Gi from interacting with GPCRs, locking Gi in its inactive GDP-bound state and blocking its ability to inhibit adenylyl cyclase, leading to unchecked cAMP elevation.
- **Option C (Incorrect):** Incorrect. Pertussis toxin specifically targets Gi, not Gq.
- **Option D (Incorrect):** Incorrect. Modification is ADP-ribosylation, not deacetylation.

Question 50

Cell Signaling

Receptor Tyrosine Kinases (RTKs), such as the Epidermal Growth Factor Receptor (EGFR), undergo which essential step upon ligand binding to initiate signaling?

- (A) Proteolytic cleavage by gamma-secretase
- (B) Receptor dimerization and trans-autophosphorylation ✓ [Correct Answer]**
- (C) G-protein coupling and GTP binding
- (D) Internalization and immediate proteasomal destruction

CORRECT ANSWER: (B) Receptor dimerization and trans-autophosphorylation**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Gamma-secretase cleaves Notch or APP, not initial RTK activation.
- **Option B (CORRECT):** Correct. Ligand binding to extracellular domains induces **RTK dimerization**. Dimerized kinase domains undergo **trans-autophosphorylation** on specific intracellular tyrosine residues, creating high-affinity docking sites for SH2-domain-containing signaling proteins.
- **Option C (Incorrect):** Incorrect. RTKs do not couple to heterotrimeric G-proteins.
- **Option D (Incorrect):** Incorrect. Autophosphorylation activates downstream cascades prior to endocytic downregulation.

Question 51

Cell Signaling

Phosphotyrosine residues on activated RTKs serve as specific binding sites for signaling proteins containing which structural protein domain?

- (A) SH2 (Src Homology 2) domain ✓ [Correct Answer]**
- (B) PH (Pleckstrin Homology) domain
- (C) Zinc finger domain
- (D) EF-hand domain

CORRECT ANSWER: (A) SH2 (Src Homology 2) domain**Detailed Choice-by-Choice Rationales:**

- **Option A (CORRECT):** Correct. **SH2 (Src Homology 2)** domains (and PTB domains) specifically recognize and bind to phosphorylated tyrosine residues within defined amino acid motifs on activated RTKs and adaptor proteins (e.g., Grb2, STATs).
- **Option B (Incorrect):** Incorrect. PH domains bind specifically to membrane phosphoinositides like PIP3.
- **Option C (Incorrect):** Incorrect. Zinc finger domains bind specific DNA sequences in nuclear transcription factors.
- **Option D (Incorrect):** Incorrect. EF-hand domains (e.g., in calmodulin) bind Ca²⁺ ions.

Question 52

Cell Signaling

In the canonical Ras-MAPK signaling pathway, what is the function of the SOS (Son of Sevenless) protein recruited by adaptor Grb2?

- (A) GTPase Activating Protein (GAP) for Ras
- (B) Guanine Nucleotide Exchange Factor (GEF) for Ras ✓ [Correct Answer]**
- (C) Serine/threonine kinase that phosphorylates ERK
- (D) Scaffold protein that degrades MEK

CORRECT ANSWER: (B) Guanine Nucleotide Exchange Factor (GEF) for Ras**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Ras-GAPs stimulate Ras GTP hydrolysis to turn Ras OFF.
- **Option B (CORRECT):** Correct. **SOS** is a **Guanine Nucleotide Exchange Factor (GEF)**. Recruited to the membrane by adaptor protein Grb2 (bound to phosphotyrosines), SOS promotes GDP dissociation and GTP loading on monomeric GTPase **Ras**, converting inactive Ras-GDP into active **Ras-GTP**.
- **Option C (Incorrect):** Incorrect. MEK is the dual-specificity kinase phosphorylating ERK.
- **Option D (Incorrect):** Incorrect. SOS acts as a GEF for Ras, not a degradation scaffold.

Question 53

Cell Signaling

Active Ras-GTP initiates a downstream serine/threonine protein kinase cascade. What is the correct sequential order of kinases in the MAPK cascade?

- (A) MEK -> Raf -> ERK
- (B) Raf -> MEK -> ERK ✓ [Correct Answer]**
- (C) ERK -> MEK -> Raf
- (D) Akt -> Raf -> MEK

CORRECT ANSWER: (B) Raf -> MEK -> ERK**Detailed Choice-by-Choice Rationales:**

- **Option A (Incorrect):** Incorrect. Raf is MAPKKK and acts upstream of MEK (MAPKK).
- **Option B (CORRECT):** Correct. Active Ras-GTP recruits and activates **Raf** (MAPK Kinase Kinase / MAP3K). Raf phosphorylates and activates **MEK** (MAPK Kinase / MAP2K). MEK dual-phosphorylates **ERK** (MAPK), which translocates to the nucleus to regulate gene expression.
- **Option C (Incorrect):** Incorrect. ERK is the terminal effector kinase in this cascade.
- **Option D (Incorrect):** Incorrect. Akt is part of the PI3K survival pathway, not the Ras-MAPK cascade.

Question 54

Cell Signaling

Phosphoinositide 3-Kinase (PI3K) activated by insulin or growth factors phosphorylates PIP2 in the plasma membrane to generate which lipid second messenger?

(A) IP3 (Inositol 1,4,5-trisphosphate)

(B) PIP3 (Phosphatidylinositol 3,4,5-trisphosphate) ✓ [Correct Answer]

(C) DAG (Diacylglycerol)

(D) Ceramide-1-phosphate

CORRECT ANSWER: (B) PIP3 (Phosphatidylinositol 3,4,5-trisphosphate)

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. IP3 is soluble inositol trisphosphate cleaved from PIP2 by PLC.
- **Option B (CORRECT):** Correct. **PI3K** phosphorylates the 3'-position of the inositol ring of PIP2 to produce **phosphatidylinositol 3,4,5-trisphosphate (PIP3)**. PIP3 serves as a membrane docking platform for Pleckstrin Homology (PH) domain proteins PDK1 and Akt (PKB).
- **Option C (Incorrect):** Incorrect. DAG is produced alongside IP3 by phospholipase C hydrolysis.
- **Option D (Incorrect):** Incorrect. Ceramide-1-phosphate is produced by ceramide kinase.

Question 55

Cell Signaling

Which lipid phosphatase acts as a major tumor suppressor by dephosphorylating PIP3 back into PIP2, thereby antagonizing PI3K/Akt pro-survival signaling?

(A) PTEN (Phosphatase and Tensin Homolog) ✓ [Correct Answer]

(B) Protein Phosphatase 1 (PP1)

(C) Calcineurin (PP2B)

(D) SHP2

CORRECT ANSWER: (A) PTEN (Phosphatase and Tensin Homolog)

Detailed Choice-by-Choice Rationales:

- **Option A (CORRECT):** Correct. **PTEN** is a 3-phosphatase tumor suppressor enzyme that specifically dephosphorylates PIP3 to PIP2. Loss-of-function mutations in PTEN cause hyperactivation of PIP3/Akt signaling, driving cell proliferation and oncogenesis in many human cancers.
- **Option B (Incorrect):** Incorrect. PP1 dephosphorylates protein serine/threonine residues (e.g., glycogen synthase).
- **Option C (Incorrect):** Incorrect. Calcineurin is a Ca²⁺/calmodulin-dependent protein phosphatase target of cyclosporine.
- **Option D (Incorrect):** Incorrect. SHP2 is a protein tyrosine phosphatase.

Question 56

Cell Signaling

The Insulin Receptor is a preformed tetrameric RTK consisting of which subunit arrangement?

- (A) Four identical alpha subunits (alpha4)
- (B) Two extracellular insulin-binding alpha subunits and two transmembrane catalytic beta subunits (alpha2-beta2) ✓ [Correct Answer]**
- (C) Single polypeptide chain with seven transmembrane helices
- (D) Dimeric beta subunits linked to GPCRs

CORRECT ANSWER: (B) Two extracellular insulin-binding alpha subunits and two transmembrane catalytic beta subunits (alpha2-beta2)

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. The receptor is a heterotetramer of distinct alpha and beta subunits.
- **Option B (CORRECT):** Correct. The **Insulin Receptor** is a disulfide-linked **alpha2-beta2 heterotetramer**. The two **alpha subunits** are entirely extracellular and bind insulin; the two **beta subunits** possess a transmembrane segment and cytosolic tyrosine kinase domain that autophosphorylates upon insulin binding.
- **Option C (Incorrect):** Incorrect. 7-TM receptors are GPCRs.
- **Option D (Incorrect):** Incorrect. Insulin receptor is an RTK, not connected to GPCRs.

Question 57

Cell Signaling

Cytokine receptors (such as Erythropoietin or Growth Hormone receptors) lack intrinsic kinase activity. Upon ligand binding, which non-receptor tyrosine kinase family do they recruit and activate?

- (A) Janus Kinases (JAKs) ✓ [Correct Answer]**
- (B) Protein Kinase A (PKA)
- (C) Mitogen-Activated Protein Kinases (MAPKs)
- (D) Glycogen Synthase Kinase 3 (GSK3)

CORRECT ANSWER: (A) Janus Kinases (JAKs)

Detailed Choice-by-Choice Rationales:

- **Option A (CORRECT):** Correct. Cytokine receptors lack intrinsic enzymatic activity. Ligand binding induces receptor dimerization, which brings associated **Janus Kinases (JAK1, JAK2, TYK2)** into proximity to cross-phosphorylate each other and the receptor tails, initiating the JAK-STAT pathway.
- **Option B (Incorrect):** Incorrect. PKA is downstream of GPCR/cAMP signaling.
- **Option C (Incorrect):** Incorrect. MAPKs are downstream of Ras/Raf.
- **Option D (Incorrect):** Incorrect. GSK3 is regulated downstream of Akt/insulin signaling.

Question 58

Cell Signaling

Upon phosphorylation by active JAKs, STAT (Signal Transducer and Activator of Transcription) proteins undergo what structural change to enter the nucleus?

(A) SH2 domain-mediated dimerization ✓ [Correct Answer]

(B) Cleavage by proteasomes

(C) Binding to GPCR beta-gamma subunits

(D) Degradation into small peptides

CORRECT ANSWER: (A) SH2 domain-mediated dimerization

Detailed Choice-by-Choice Rationales:

- **Option A (CORRECT):** Correct. JAKs phosphorylate specific tyrosine residues on cytokine receptors. STAT proteins dock via their SH2 domains, get phosphorylated by JAKs, dissociate from the receptor, and form ****reciprocal SH2-phosphotyrosine dimers****. These STAT dimers translocate directly to the nucleus to induce gene transcription.
- **Option B (Incorrect):** Incorrect. Phosphorylation activates STAT dimerization, not proteasomal destruction.
- **Option C (Incorrect):** Incorrect. STATs operate independently of GPCR subunits.
- **Option D (Incorrect):** Incorrect. STAT dimers are functional nuclear transcription factors.

Question 59

Cell Signaling

Steroid hormones (such as cortisol, estrogen, and testosterone) cross the plasma membrane and bind to intracellular nuclear receptors. Which structural motif on these receptors mediates binding to specific Hormone Response Elements (HREs) on DNA?

(A) Leucine zipper

(B) Zinc finger domain ✓ [Correct Answer]

(C) SH3 domain

(D) Pleckstrin homology domain

CORRECT ANSWER: (B) Zinc finger domain

Detailed Choice-by-Choice Rationales:

- **Option A (Incorrect):** Incorrect. Leucine zippers mediate transcription factor dimerization (e.g., Jun/Fos).
- **Option B (CORRECT):** Correct. Nuclear hormone receptors possess a central, highly conserved ****DNA-Binding Domain (DBD)**** containing two ****Zinc Finger motifs**** (each coordinating a Zn²⁺ ion with 4 cysteine residues). These zinc fingers insert into the major groove of DNA to recognize specific Hormone Response Elements (HREs).
- **Option C (Incorrect):** Incorrect. SH3 domains bind proline-rich peptides in cytosolic signaling.
- **Option D (Incorrect):** Incorrect. PH domains bind membrane phosphoinositides.

Question 60

Cell Signaling

In the intrinsic (mitochondrial) pathway of apoptosis, release of Cytochrome c from the mitochondrial intermembrane space into the cytosol induces assembly of the 'Apoptosome' complex by binding to which cytosolic protein?

(A) Apaf-1 (Apoptotic Protease Activating Factor-1) and procaspase-9 ✓ [Correct Answer]

(B) FADD and procaspase-8

(C) Bcl-2 and Bax

(D) TNF receptor and TRADD

CORRECT ANSWER: (A) Apaf-1 (Apoptotic Protease Activating Factor-1) and procaspase-9

Detailed Choice-by-Choice Rationales:

- **Option A (CORRECT):** Correct. Cytochrome c released into cytosol binds to **Apaf-1**. In the presence of dATP/ATP, Apaf-1 oligomerizes into a heptameric wheel structure called the **Apoptosome**, which recruits and auto-activates initiator **Procaspase-9**. Active Caspase-9 then cleaves executioner caspases (Caspase-3 and -7).
- **Option B (Incorrect):** Incorrect. FADD and Procaspase-8 assemble the Death-Inducing Signaling Complex (DISC) in the **extrinsic** death receptor pathway.
- **Option C (Incorrect):** Incorrect. Bcl-2 (anti-apoptotic) and Bax (pro-apoptotic) regulate mitochondrial membrane permeability upstream of cytochrome c release.
- **Option D (Incorrect):** Incorrect. TNF receptor and TRADD initiate the extrinsic TNF-alpha apoptotic pathway.